

MATH9102

HYPOTHESIS TESTING

Sources used in creation of this lecture:

Statistics and Data Analysis, Peck, Olsen and Devore; Discovering Statistics Using IBM SPSS, Andy Field; Understanding Basic Statistics, Brase and Brase; SPSS Survival Manual, Julie Pallant

Statistical Power

TYPE I AND TYPE II ERRORS

Errors in Hypothesis Tests

Just because we find a statistically significant difference/effect does not necessarily indicate there is a causal relationship.

Because the hypothesis test relies on sample data, and because sample data are not completely reliable, there is always the risk that misleading data will cause the hypothesis test to reach a wrong conclusion.

Power of a Hypothesis Test

The **power** β of a hypothesis test is defined as the probability that the test will reject the null hypothesis when there is no effect.

The power of a test depends on a variety of factors including the size of the effect and the size of the sample.

Type I Errors

When the sample data appear to show an effect/difference when, in fact, there is none in the population.

- In this case the researcher will reject the null hypothesis and falsely conclude that there is an effect/difference.

Caused by unusual, unrepresentative samples.

- E.g. The researcher selects an extreme sample with the result that the sample falls in the critical region even though there is no effect.

The hypothesis test is structured so that Type I errors are very unlikely

- Specifically, the probability of a Type I error is equal to the **alpha level (α)** – also called cut-off level

Type II Errors

When the sample does not appear to have an effect/difference when in fact this exists in the population.

- In this case, the researcher will fail to reject the null hypothesis.
- Commonly where you are looking small effects/differences (not large enough to show up in your data).

Possible Error...?

Compare Type I and Type II error like this:

- The only concern when you find statistical significance ($p < 0.05$) is Type I Error
 - Is the difference between groups REAL or due to Random Sampling Error
 - Thankfully, the p-value tells you exactly what the probability of that random sampling error is
 - In other words, the p-value tells you how likely Type I error is given your alpha level

Possible Error...?

Does the p-value tell you how likely Type II error is?

- The probability of Type II error is better provided by **Power**

Possible Error...?

Statistical Power, also known as β (Beta)
(actually $1 - \beta$)

Power (β) is related to Alpha (α)

But:

- Alpha is the probability of having Type I error
 - Lower number is better (i.e., 0.05 vs 0.01 vs 0.001)
- Power is the probability of **NOT** having Type II error
 - The probability of being right (correctly rejecting the null hypothesis)
 - Higher number is better (typical goal is 0.80)

Should it be statistically significant?

The most obvious thing you need to consider is if you REALLY should have found a statistically significant result?

- Just because you wanted your test to be significant doesn't mean it should be
- This wouldn't be Type II error – it would just be the correct decision

Power analysis

An important aspect of experimental design.

Allows us to determine the sample size required to detect an effect of a given size with a given degree of confidence.

- Also allows us to determine the probability of detecting an effect of a given size with a given level of confidence, under sample size constraints.
- If the probability is unacceptably low, we would be wise to alter or abandon the experiment.

Power Levels

Most fields conventionally choose: **80% power** (0.80)

- Sometimes **90%** in high-stakes research (e.g., clinical trials)

Power balances:

- The risk of **Type II error** (missing a real effect)
- Practical limits (cost, time, feasibility)

80% power means:

- 20% chance of failing to detect a real effect
- Widely accepted as a reasonable trade-off

Power Levels

Power	What happens if the effect exists?	Interpretation
80%	You detect it 4 out of 5 times	Acceptable but still 1 in 5 chance you'd miss it
90%	You detect it 9 out of 10 times	Much lower risk of overlooking it

Power Analysis

1. sample size
2. effect size
3. significance level
 - $P(\text{Type I error}) = \text{probability of finding an effect that is not there}$
4. power
 - *Probability of finding an effect that is truly there*
 $P(\text{Type II error}) = \text{Probability of *not* finding an effect that is truly there}$
 - Given any three, we can determine the fourth.

Power Analysis

The `pwr` package implements power analysis as outlined by Cohen

Function	power calculations for
<code>pwr.2p.test</code>	two proportions (equal n)
<code>pwr.2p2n.test</code>	two proportions (unequal n)
<code>pwr.anova.test</code>	balanced one way ANOVA
<code>pwr.chisq.test</code>	chi-square test
<code>pwr.f2.test</code>	general linear model
<code>pwr.p.test</code>	proportion (one sample)
<code>pwr.r.test</code>	correlation
<code>pwr.t.test</code>	t-tests (one sample, 2 sample, paired)
<code>pwr.t2n.test</code>	t-test (two samples with unequal n)

Power Analysis

t-test

```
pwr.t.test(n = , d = , sig.level = ,  
power = , type = c("two.sample",  
"one.sample", "paired"))
```

where n is the sample size

d is the effect size,

and type indicates a two-sample t-test, one-sample t-test or paired t-test. If you have unequal sample sizes, use

E.g.

What size sample is needed to detect an effect size of 0.8 at the significance level 0.01 with a power of 0.8

```
pwr.t.test(d=0.8, sig.level=.01, power  
=0.8, alternative="two.sided")
```

Power Analysis

For a one-way analysis of variance use

```
pwr.anova.test(k = , n = , f =  
, sig.level = , power = )
```

where k is the number of groups and n is the common sample size in each group.

For a one-way ANOVA effect size is measured by f where

Cohen suggests that f values of 0.1, 0.25, and 0.4 represent small, medium, and large effect sizes respectively.

Online Calculators

A range of online calculators are available at

- <http://statpages.info/#Power>

You can also use GPower

- A downloadable application - <http://www.gpower.hhu.de/>

How can I increase Power?

Increase Alpha level

- Changing alpha from 0.05 to 0.10 will increase your power (better chance of finding significant results)
- Downsides to increasing your alpha level?
 - This will increase the chance of Type I error
 - This is rarely acceptable in practice
- Only really an option when working in a new area:
 - Researchers are unsure of how to measure a new variable
 - Researchers are unaware of confounders to control for

How can I increase Power?

Increase N (sample size)

- Sample size is directly used when calculating p-values
- Including more subjects will increase your chance of finding statistically significant results
- Downsides to increasing sample size?
 - More subjects means more time/money
- More subjects is ALWAYS a better option if possible

How can I increase Power?

Use fewer groups/variables (simpler designs)

- Related to sample size but different
 - 'Use fewer groups' NOT 'Use less subjects'
- ↑ groups negatively effects your degrees of freedom
 - Remember, df is calculated with # groups and # subjects
- Lots of variables, groups and interactions make it more difficult to find statistically significant differences
 - The purpose of the Family-wise error rate is to make it harder to find significant results!
- Downsides to fewer groups/variables?
 - Sometimes you NEED to make several comparisons and test for interactions - unavoidable

How can I increase Power?

Measure variables more accurately

- If variables are poorly measured (sloppy work, broken equipment, outdated equipment, etc...) this increases measurement error
- More measurement error decreases confidence in the result
- More of an internal validity problem than statistical problem
- Downsides to measuring more accurately?
 - None – if you can afford the best tools

How can I increase Power?

Decrease subject variability

- Subjects will have various characteristics that may also be correlated with your variables
 - E.g. for research involving people gender, race/ethnicity, age, etc...
 - These variables can confound your results, making it harder to find statistically significant results
 - When planning your sample (to enhance power), select subjects that are very similar to each other
 - This is a reason why repeated measures tests and paired samples are more likely to have statistically significant results
- Downside to decreasing subject variability?
 - Will decrease your external validity – generalizability
 - If you only test women, your results do not apply to men

How can I increase Power?

Increase magnitude of the mean difference

- If your groups are not different enough, make them more different
 - Compare a 'very' high group to a 'very' low group
 - Sampling at the extremes, getting rid of the middle group
- Downsides to using the extremes?
 - Similar to decreasing subject variability, this will decrease your external validity

The Catch-22 of Power and P-values

The larger your sample, the more likely you'll find statistically significant results

- Sometimes miniscule differences between groups or tiny correlations are 'significant'
- This becomes relevant once sample size grows to 100~150 subjects per group
- Once you approach 1000 subjects, it's hard not to find $p < 0.05$

Example from most highly cited paper in psychology, 2004...

[\[HTML\] The effects of violent video game habits on adolescent hostility, aggressive behaviors, and school performance](#)

[\[HTML\] sciencedirect.com](#)

[DA Gentile](#), [PJ Lynch](#), [JR Linder](#), [DA Walsh](#) - [Journal of adolescence](#), 2004 - Elsevier

... term, trait **hostility** is a mediator of the **effects** of **violent video games** on **aggressive** behavior.

... mediating **effect** of **hostility**, in which **hostility** mediates the **effects** of both amount of play and ...

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Paper Citation

Gentile, D. A., Lynch, P. J., Linder, J. R., & Walsh, D. A. (2004). The effects of violent video game habits on adolescent hostility, aggressive behaviors, and school performance. *Journal of adolescence*, 27(1), 5-22.

This paper was the first to report finding a link between playing video games/TV and aggression in children.

The effects of violent video game habits on adolescent hostility, aggressive behaviors, and school performance

Douglas A. Gentile^{a,*}, Paul J. Lynch^b, Jennifer Ruh Linder^c, David A. Walsh^a

^a*National Institute on Media and the Family, 606 24th Avenue South, Suite 606, Minneapolis, MN 55454, USA*

^b*University of Oklahoma Medical School, USA*

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Table 4

Correlations between media habits and parental limits and outcomes ($n = 399\text{--}586$)

	Trait hostility	Arguments with teachers	Physical fights	Grades
Amount variables				
Amount of video game play	0.20***	0.12*	0.21***	-0.25***
Amount of time watching TV	0.20***	0.10*	0.12***	-0.20***
Amount of reading for pleasure	-0.08 [†]	-0.17***	-0.07	0.07 [†]
Violent content variables				
Violent video game exposure	0.21***	0.20***	0.32***	-0.23***
Preferred violence in video games	0.31***	0.25***	0.36***	-0.34***
Preferred violence compared to 2 or 3 years ago	0.23***	0.16**	0.19***	-0.14**
Parental involvement scale	-0.14**	-0.27***	-0.18***	0.27***

[†] $p < 0.09$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Degrees of freedom

- Different numbers used for each calculation – but which calculation used which n ?

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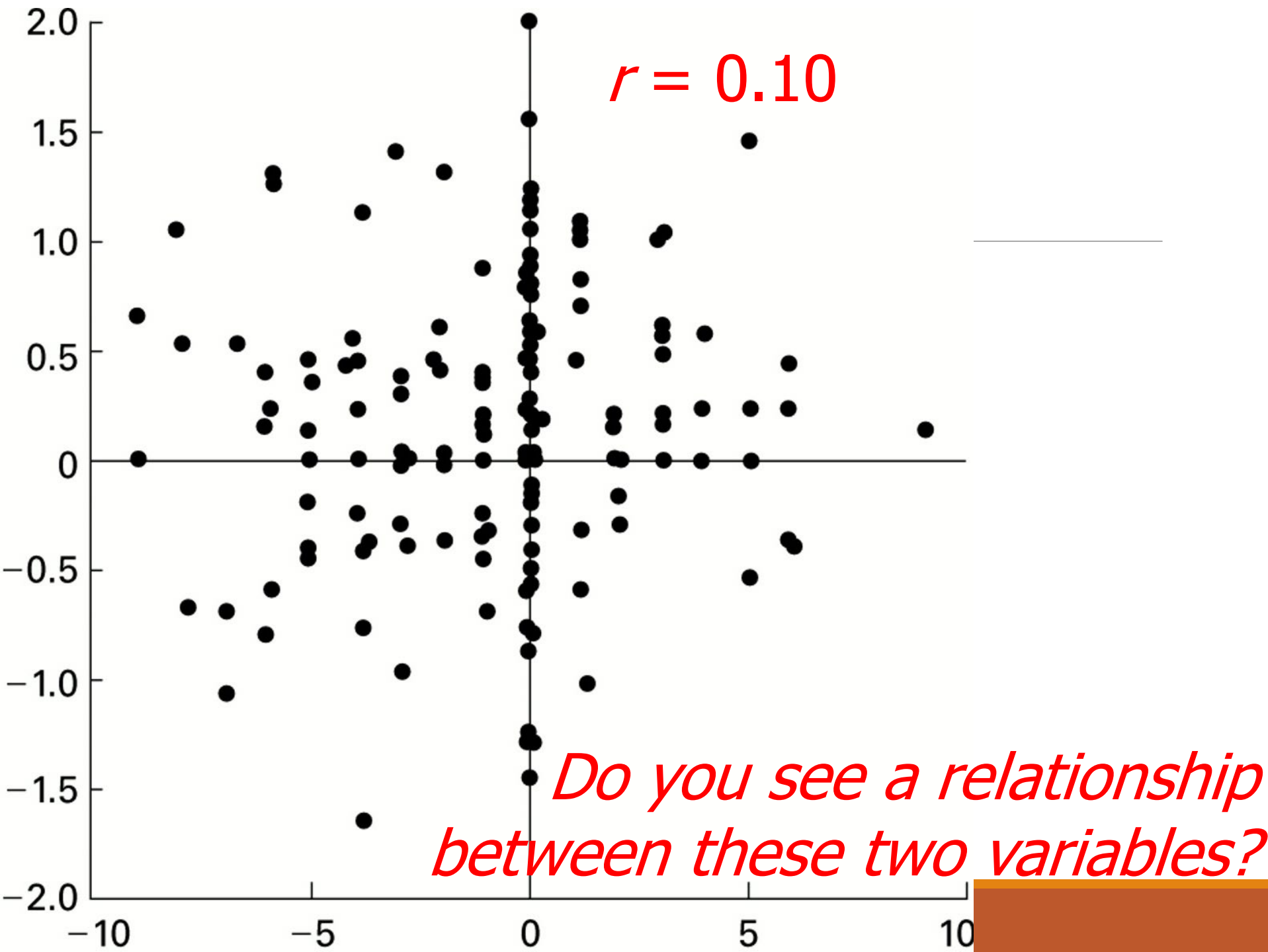
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How strong are these correlations?

What does a correlation of 0.10 look like?



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Nearly every correlation in this table has $p < 0.05$

But how strong are these correlations?

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P-value

P<0.09?

Warning

Treat other people's statistics with care !

- Look behind the headlines

Measuring Effect Size

A hypothesis test evaluates the statistical significance of the results from a research study.

The hypothesis test is influenced not only by the size of the effect/difference but also by the size of the sample.

Thus, even a very small effect/difference can be significant if it is observed in a very large sample.

Finding a statistically significant result does not necessarily mean a large effect

It is recommended that the hypothesis test be accompanied by a measure of the **effect size**.

Cohen's measures of effect size are used as standard.

Effect Size

To get an idea of how ‘important’ a difference or association is, we can use Effect Size

- There are over 40 different types of effect size
 - Depends on statistical test used

Effect size is like a ‘descriptive’ statistic that tells you about the magnitude of the association or group difference

- Not impacted by statistical significance
- Effect size can stay the same even if p-value changes
- Present the two together when possible

Another example

If we sample a group of 100 students and find their average IQ is 103.

- The population mean for IQ is 100, SD = 15.
- We run a one-sample t-test and find it to be statistically significant ($p < 0.05$)
- However, effect size is...
 - 0.2, or Small Effect
- Interpretation: While this difference is likely not due to random sampling error – it's not very important either

Confidence Intervals

Confidence Intervals

A range of values we believe the **true population value** lies within.

- Example:
 - Mean = 50
 - 95% CI = **47 to 53**
 - → We are **95% confident** the true mean is somewhere between **47 and 53**

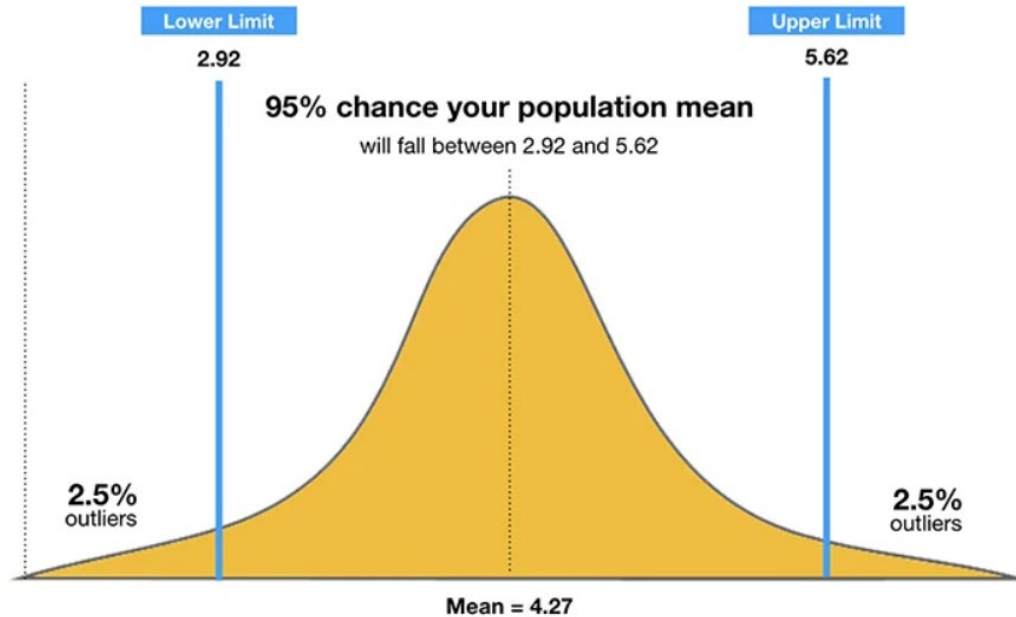
Key idea:

- It's not about the sample — it's about **what's true in the population**

Why use them?

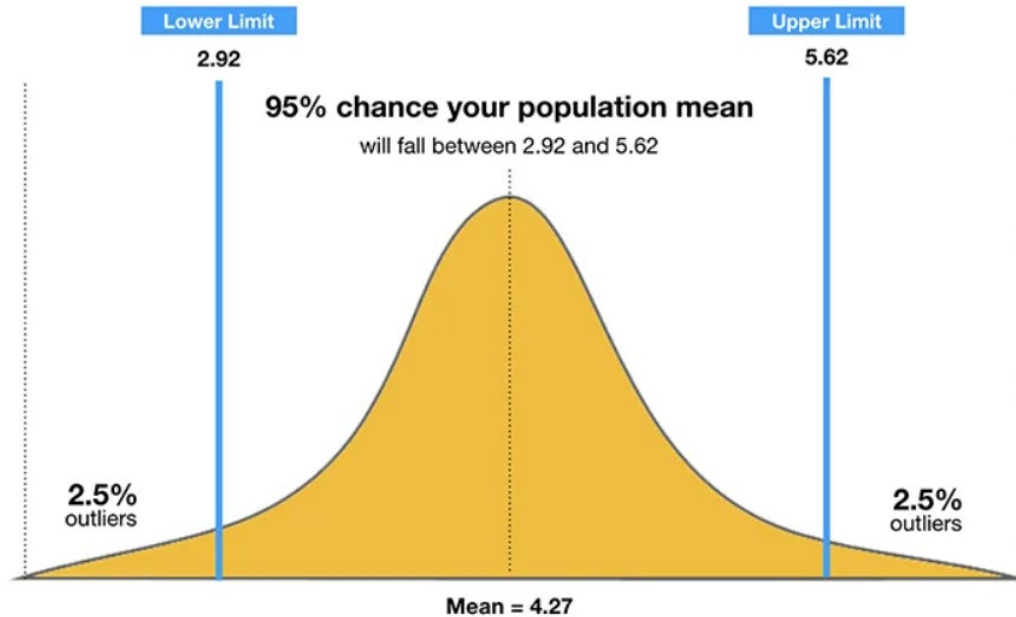
- Shows **uncertainty** in our estimate
- Shows how **precise** the results are
- More informative than just a p-value

Confidence Intervals help avoid mistakes like finding tiny effects that are “significant” just because the sample is hug



Confidence Intervals

Due to natural sampling variability, the sample mean (center of the CI) will vary from sample to sample.



Confidence Intervals

Confidence is in the method

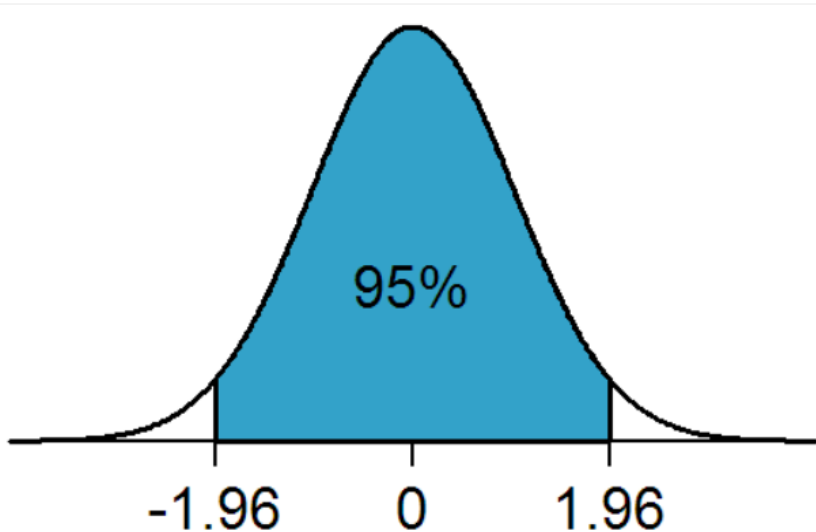
If we repeated the sampling method many times, approximately 95% of the intervals constructed would capture the true population mean.

Confidence Intervals

Normal Distribution

The probability of the population mean value being between -1.96 and $+1.96$ standard deviations (z-scores) from the sample mean is 95%

There is a 5% chance that the population mean lies outside of the upper and lower confidence interval (the 2.5% of outliers on either side of the 1.96 z-scores).



Writing Hypotheses

For Linear Regression

Null (H_0):

The predictor does **not** help predict the outcome

Alternative (H_1):

The predictor **does** help predict the outcome

In prediction, hypotheses test whether a variable **adds useful predictive information** beyond randomness.

For Linear Regression

H_0 : Study hours do not help predict exam score

H_1 : Study hours do help predict exam score

Interpretation of finding evidence to reject H_0 :

- Study hours improve prediction of exam performance.

For Linear Regression

H_0 : Study hours do not improve prediction of exam score after controlling for GPA

H_1 : Study hours do improve prediction of exam score after controlling for GPA

H_0 : Predicted exam scores do not differ by gender after controlling for study hours

H_1 : Predicted exam scores do differ by gender after controlling for study hours

H_0 : The relationship between study hours and exam score is the same for males and females controlling for GPA

H_1 : The relationship differs by gender controlling for GPA

For Logistic Regression

Hypotheses

Null (H_0):

Predictor does **not** affect odds of outcome

Alternative (H_1):

Predictor **does** affect odds

Logistic Regression

H_0 : BMI does not predict disease status

H_1 : BMI does predict disease status

H_0 : BMI does not predict disease status controlling for age

H_1 : BMI does predict disease status controlling for age

H_0 : Disease odds do not differ by gender controlling for BMI

H_1 : Disease odds do differ by gender controlling for BMI

H_0 : Disease odds do not differ by gender controlling for BMI

H_1 : Disease odds do differ by gender controlling for BMI